

# OCTONIONIC ACTION FROM THE HQIV-LEAN VARIATIONAL LAYER: DERIVATION, HOLONOMY ALIGNMENT, AND A TIERED UNIQUENESS THESIS

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ABSTRACT. This manuscript is a variational sequel to [1] and [2]. Those works package discrete null-shell growth, a normalized cumulative phase readout  $\mathcal{R}$ , an explicit phase-lift generator  $\Delta$ , and machine-checked Lie closure  $G_2 \cup \{\Delta\} \Rightarrow \mathfrak{so}(8)$  on the HQVM octonionic seed. Here we center the *action* layer already present in the HQIV-LEAN library: a discrete octonion-indexed gauge potential  $A^a{}_\mu$ , field strength  $F^a{}_{\mu\nu}$ , O–Maxwell Lagrangian, Euler–Lagrange covector, gravitational constraint action tied to the HQVM Friedmann identity, and the holonomy glue that re-packages the *same*  $F$  data as abelian plaquette transports with a proved cyclic Stokes identity. The holonomy bridge matters because it ties the variational kinetic aggregate to Wilson-type defects on the minimal Fin 4 cycle without changing the underlying discrete flux—the same bookkeeping the rapidity layer uses before Lie closure.

We state with file-level precision which claims are *proved in Lean* (Euler–Lagrange identities matching the discrete divergence of  $F$ , cyclic flux sum zero, two-sided bounds comparing  $L_O$  kinetic to cyclic Wilson squares, Friedmann constraint as  $S_{\text{grav}} = 0$ ) and which steps remain narrative or scaffolded (continuum limit of  $\mathcal{R}$  to connection data, manifold integration, path-integral measure). The *tiered uniqueness thesis* is the paper’s conceptual payoff: (I) within the declared polynomial gauge ansatz, the EL slot is rigid—the discrete implementation is the exact finite-difference analogue of the continuum Maxwell pattern; (II) Hurwitz rigidity and eight-dimensional division-algebra uniqueness align the Fin 8 carrier with triality/Spin(8) and the certified  $G_2 \cup \{\Delta\}$  seed; (III) a structured Tier III program (continuum limit with error control, paired Einstein/ $G_2$  stationarity, associator/measure) replaces a single vague conjecture. The goal is to advance the action story on its own audit spine without folding it into the shell-combinatorics graph of the predecessors. For readers coming from lattice gauge theory, the holonomy lemmas position the same  $F$  slots where Wilson plaquette language applies [4, 6]. We also name *lapse dragging*: the HQVM identification of the cumulative lapse increment  $\phi t/c$  above the Newtonian  $\Phi$  piece with the *same* auxiliary scalar that enters  $\mathcal{L}_\phi$  and  $G_{\text{eff}}(\phi) = \phi^{3/5}$  (HQVM\_lapse, same\_phi\_in\_O\_Maxwell\_and\_HQVM in GRFromMaxwell.lean); Remark 3.4 records sample  $G_{\text{eff}}/G_0$  ratios and links that single- $\alpha$  pipeline to galactic rotation-curve phenomenology in the Jacobson/Brodie horizon picture [5, 9].

## 1. LOGICAL PLACEMENT AND READER CONTRACT

### 1.1. Predecessors. [1] gives the chain

$$\text{uniform growth} \implies K \implies \Omega \implies \mathcal{R} \implies \Delta \implies \langle G_2, \Delta \rangle_{\text{Lie}} = \mathfrak{so}(8)$$

under explicit imports (quadratic shell law,  $\alpha = 3/5$  normalization story, Fano octonion model,  $(e_1, e_7)$  plane). [2] isolates the quadratic-versus-cubic dimensional hinge in 3+1, records conditional forcing theorems in `Hqiv/Story/CausalRapidityForcing.lean`, and develops Hurwitz/triality narrative with careful scope.

**1.2. Where this paper sits in the three-layer stack.** It is useful to picture HQIV formalization as three layers that *compose* in the repository narrative but have different logical strengths. The **discrete growth layer** (shell counts,  $K, \Omega, \mathcal{R}$ , lattice- $\alpha$  coherence) is the strongest combinatorial spine; the **Lie closure layer** ( $G_2 \cup \{\Delta\} \Rightarrow \mathfrak{so}(8)$  on explicit matrices) is a certified linear-algebra certificate once seeds are fixed; the **variational layer** of the present manuscript (Hqiv/Physics/Action.lean and holonomy glue) packages equations of motion and constraints as stationarity of explicit functionals over the same octonion-indexed cutoff. This paper does *not* merge those layers into a single implication from poset axioms; it documents the third layer precisely and explains how it is designed to interoperate with the first two through shared parameters  $(\alpha, \phi)$ .

**1.3. What this paper adds (Lean modules).**

- **Discrete O–Maxwell action and EL identities** (Hqiv/Physics/Action.lean; Sect. 2). Definitions `F_from_A`, `L_O_kinetic`, `L_O_Maxwell_general`, `EL_O_general`, theorems `action_O_Maxwell_EL_eq_emergent_general`, `superposition action_O_Maxwell_general_add_J`.
- **Gravity constraint and total action** (Hqiv/Physics/Action.lean; Sect. 3). `S_HQVM_grav`, `S_HQVM_grav_zero_iff_Friedmann`, `action_total_general`, `action_total_general_add_J`, `equations_from_action`.
- **Holonomy alignment on the same  $F$  slots** (Hqiv/Physics/ActionHolonomyGlue.lean; Sect. 4). `sum_F_cyclicIndex_eq_zero`, `discreteSquareHolonomy_F_cyclic_eq_one`, `L_O_kinetic_le_neg_quarter_sum_cyclic_wilson_sq`, `L_O_kinetic_ge_neg_half_sum_cyclic_wilson_sq`, `L_O_kinetic_two_sided_cyclic_wilson_sq`.

**1.4. Scope and not claimed.**

- Causal poset axioms *alone* do not imply this action; the quadratic shell law, octonion table, and plane choices remain explicit imports where used.
- No continuum path integral, anomaly cancellation, or uniqueness of measure is proved here.
- No claim that alternative discrete variational ansätze cannot be written off the HQIV design path; Tier I uniqueness is *internal* to the stated Lagrangian class.
- The functorial bridge  $\mathcal{R} \rightsquigarrow A$  with analytic error control is deferred to Hqiv/Geometry/ManifoldLagrangianScaffold.lean and LatticeContinuumActionCoincidence (Sect. 5).

## 2. DISCRETE O–MAXWELL ACTION

Fix octonion channel index  $a \in \text{Fin } 8$  and spacetime index  $\mu, \nu \in \text{Fin } 4$  as in the library.

**Definition 2.1** (Discrete field strength (Lean: `F_from_A`)). *For  $A : \text{Fin } 8 \rightarrow \text{Fin } 4 \rightarrow \mathbb{R}$ ,*

$$F^a_{\mu\nu}(A) := A^a_\nu - A^a_\mu.$$

**Definition 2.2** (Kinetic and source densities (Lean)). *The kinetic aggregate is (`L_O_kinetic`)*

$$(1) \quad \mathcal{L}_{\text{kin}}(A) := -\frac{1}{4} \sum_{a \in \text{Fin } 8} \sum_{\mu, \nu \in \text{Fin } 4} \frac{(F^a_{\mu\nu}(A))^2}{2}.$$

*The factor  $\frac{1}{2}$  inside the sum implements antisymmetric bookkeeping while summing over all ordered pairs  $(\mu, \nu)$ . The source interaction for  $J_{\text{src}} : \text{Fin } 8 \rightarrow \text{Fin } 4 \rightarrow \mathbb{R}$  is (`L_O_source_general`)*

$$(2) \quad \mathcal{L}_{\text{src}}(J_{\text{src}}, A) := \sum_{a \in \text{Fin } 8} \sum_{\nu \in \text{Fin } 4} (J_{\text{src}})^a_\nu A^a_\nu.$$

**Definition 2.3** (Full O–Maxwell density and action cell (Lean)). *For  $\phi_{\text{val}} > -1$ , the  $\phi$ -coupling term is (`L_O_phi_coupling`)*

$$\mathcal{L}_\phi(A, \phi_{\text{val}}) := \alpha \log(\phi_{\text{val}} + 1) \sum_{\nu \in \text{Fin } 4} (\text{grad}_\phi)_\nu A^0_\nu,$$

*with  $(\text{grad}_\phi)_\nu = \text{grad\_phi } \nu$  (currently the zero placeholder in `Action.lean`; continuum gradients use `ContinuumOmaxwellClosure`). The full density is*

$$\mathcal{L}_O[J_{\text{src}}](A, \phi_{\text{val}}) := \mathcal{L}_{\text{kin}}(A) + 4\pi \mathcal{L}_{\text{src}}(J_{\text{src}}, A) + \mathcal{L}_\phi(A, \phi_{\text{val}}),$$

*(`L_O_Maxwell_general`). The action cell is the same real number (`action_O_Maxwell_general`).*

**Theorem 2.4** (Discrete Euler–Lagrange matches divergence of  $F$  minus sources (Lean)). *For  $J_{\text{src}}, A, \phi_{\text{val}}$  with  $\phi_{\text{val}} + 1 > 0$ , the component `EL_O_general` at  $(a, \nu)$  equals*

$$\sum_{\mu \in \text{Fin } 4} F^a_{\mu\nu}(A) - 4\pi (J_{\text{src}})^a_\nu - \begin{cases} \alpha \log(\phi_{\text{val}} + 1) (\text{grad}_\phi)_\nu, & a = 0, \\ 0, & a \neq 0. \end{cases}$$

*This is `action_O_Maxwell_EL_eq_emergent_general` (and `action_O_Maxwell_EL_eq_emergent` for `J_O`).*

*Sketch (conceptual; Lean is definitional/rw). Vary  $A^b_\sigma \rightarrow A^b_\sigma + \varepsilon \delta^b_a \delta^\sigma_\nu$  in  $\mathcal{L}_{\text{kin}}$ . Each  $F^b_{\mu\sigma}$  shifts by  $\varepsilon(\delta^\mu_\sigma - \delta^\mu_\nu)$  when  $\sigma = \nu$ , and each  $F^b_{\nu\mu}$  shifts by the opposite increment. Collecting coefficients of  $\varepsilon$  produces a telescoping sum over  $\mu$  that yields  $\sum_\mu F^a_{\mu\nu}$ —the discrete analogue of  $\partial_\mu F^{\mu\nu}$ . The source term  $4\pi J \cdot A$  contributes  $-4\pi J^a_\nu$  to the EL covector, and the  $\phi$ -coupling contributes only on channel  $a = 0$  by construction. This is the standard polynomial-gauge calculation carried out on the finite index set.  $\square$*

**Remark 2.5** ( $\alpha$ ,  $\gamma_{\text{HQIV}}$ , and the AuxiliaryField ladder). *The imprint  $\alpha = 3/5$  enters  $\mathcal{L}_\phi$  explicitly. The same  $\alpha$  is tied to the null-shell curvature density in `OctonionicLightCone` (`shell_shape`, `alpha_eq_3_5`); readers who arrive from the growth manuscript and want the thermodynamic packaging of the imprint fractions (Jacobson/Brodie on horizons, monogamy template) should consult the appendix Thermodynamic Derivation of the  $(\alpha_d, \gamma_d)$  Family in [2] (Lean anchor `sec:thermo-derivation` in `papers/hqiv_3d_causal_growth_octonionic_gauge.tex`). The auxiliary scalar follows the temperature ladder in `Hqiv/Geometry/AuxiliaryField.lean`: `phi_of_shell` at shell index  $m$  equals `phiTemperatureCoeff` divided by  $T(m)$ , with `phiTemperatureCoeff` the proved constant 2 (`phiTemperatureCoeff_eq_two`), and `shell_shape_in_terms_of_phi` rewrites the shell readout using  $\phi(m)/2 = m+1$ . On the gravity side, `gamma_HQIV` enters the Friedmann coefficient  $(3 - \gamma_{\text{HQIV}})$  in `S_HQVM_grav`; Lean proves  $(3 - \gamma_{\text{HQIV}}) = 13/5$  (`three_minus_gamma_eq` in `Hqiv/Geometry/HQVMetric.lean`). Thus  $(\alpha, \gamma)$  is simultaneously the monogamy/unit-split pair from the growth papers and the coupling/normalization pair threading `Action`, `HQVMetric`, and `AuxiliaryField`.*

**Remark 2.6** (Relation to `ModifiedMaxwell.emergentMaxwellInhomogeneous_O_general`). *`Hqiv/Physics/ModifiedMaxwell.lean` packages the emergent O-sector equation with the same  $-4\pi J_{\text{src}}$  slot and  $\alpha$ -weighted  $\phi$ -gradient correction (via `grad_phi` in `Action.lean`, the placeholder gradient in `ModifiedMaxwell.lean`, and `coordsGradientComponents` in `ContinuumOmaxwellClosure`).*

The discrete Action *EL* uses an explicit  $\sum_\mu F$  surrogate while the emergent file still carries a placeholder *O.O* for the covariant divergence term; the maintained design contract is source-slot alignment between *EL* and emergent inhomogeneous equations. Completing the identification requires wiring the same *coordsGradientComponents* into both sides—already started in *ContinuumOmaxwellClosure*.

### 3. GRAVITATIONAL SECTOR AND TOTAL ACTION

**Theorem 3.1** (Gravitational constraint equals Friedmann (Lean)). For  $\phi, \rho_m, \rho_r \in \mathbb{R}$ , with *S\_HQVM\_grav* and *HQVM\_Friedmann\_eq* as in the library,

$$S_{HQVM\_grav}(\phi, \rho_m, \rho_r) = 0 \iff HQVM\_Friedmann\_eq \phi \rho_m \rho_r.$$

This is *S\_HQVM\_grav\_zero\_iff\_Friedmann*. Equivalently, for  $\phi \geq 0$ , *O\_Maxwell\_compatible\_with\_HQVM\_GR\_homogeneous* rewrites the constraint as  $(13/5)\phi^2 = 8\pi\phi^\alpha(\rho_m + \rho_r)$ .

**Definition 3.2** (Total action and current superposition (Lean)). Write  $S_O[J_{src}](A, \phi_{val}) := \mathcal{L}_O[J_{src}](A, \phi_{val})$  for the *O-Maxwell* cell and let  $S_{grav}(\phi, \rho_m, \rho_r)$  denote *S\_HQVM\_grav* evaluated at  $(\phi, \rho_m, \rho_r)$ . The total cell is

$$S_{tot}[J_{src}](A, \phi, \rho_m, \rho_r) := S_O[J_{src}](A, \phi) + S_{grav}(\phi, \rho_m, \rho_r),$$

as defined by *action\_total\_general*. For two sources, Lean proves

$$(3) \quad S_{tot}[J_1 + J_2](A, \phi, \rho_m, \rho_r) = S_{tot}[J_1](A, \phi, \rho_m, \rho_r) + S_{tot}[J_2](A, \phi, \rho_m, \rho_r) \\ - \mathcal{L}_{kin}(A) - \mathcal{L}_\phi(A, \phi) - S_{grav}(\phi, \rho_m, \rho_r).$$

This is *action\_total\_general\_add\_J* (built from *action\_O\_Maxwell\_general\_add\_J*). The *O-Maxwell* part alone satisfies the simpler linearity in *J* only for the  $4\pi J \cdot A$  piece; the kinetic and  $\phi$  terms are shared, so naive addition of two full densities double-counts those shared blocks.

**Example 3.3** (Flat potential, trivial current, balanced densities). Take  $A^a_\mu \equiv 0$  so  $F^a_{\mu\nu} = 0$  and  $\mathcal{L}_{kin} = 0$ ; take  $J_\cdot O \equiv 0$  so  $\mathcal{L}_{src} = 0$ ; take *grad\_phi*  $\equiv 0$  so  $\mathcal{L}_\phi = 0$ . Then  $S_O = 0$ . Choose  $\phi = 1$  (Hubble reference in natural units). Friedmann requires  $(13/5)\phi^2 = 8\pi\phi^\alpha(\rho_m + \rho_r)$  with  $\alpha = 3/5$ , hence  $\rho_m + \rho_r = 13/(40\pi)$ . For these values  $S_{grav} = 0$  and *EL\_O* reduces to  $\sum_\mu F = 0$ , which holds trivially. This is the same decoupled “zero sector” used in *equations\_from\_action* for illustration: gravity is a constraint on  $(\phi, \rho)$ ; gauge *EL* is a separate stationarity problem in *A* once *J* and  $\phi$  data are fixed.

**Remark 3.4** (Lapse dragging, lapse-corrected GR, and the demystified  $\Lambda$ ). The constraint  $S_{grav} = 0$  of Theorem 3.1 is not the textbook flat FLRW equation of Einstein GR with a fundamental cosmological constant  $\Lambda$ ,

$$H^2 = \frac{8\pi G}{3} \rho_{tot} + \frac{\Lambda c^2}{3}.$$

Instead, for  $\phi \geq 0$ , Lean rewrites the HQVM Friedmann identity as (*O\_Maxwell\_compatible\_with\_HQVM\_GR\_homogeneous*)

$$(4) \quad \frac{13}{5} \phi^2 = 8\pi G_{eff}(\phi) (\rho_m + \rho_r), \quad G_{eff}(\phi) = \phi^\alpha, \quad \alpha = \frac{3}{5}$$

(*G\_eff\_eq*), where  $(3 - \gamma_{HQIV}) = 13/5$  is the unit-split/monogamy coefficient (*three\_minus\_gamma\_eq*). There is therefore **no independent  $\Lambda$  knob**: whatever plays the role of a “dark” constant term in a standard frame is packaged here into the  $\phi$ -dependent coupling  $G_{eff}$  and the fixed

rational prefactor inherited from the same  $(\alpha, \gamma)$  ledger that the thermodynamic appendix of [2] derives from Jacobson/Brodie horizon bookkeeping. In the HQVM narrative,  $\phi$  is the relative-observer energy normalizer (auxiliary field / temperature ladder), not an ad hoc fifth force; identifying  $H$  with  $\phi$  in the homogeneous sector is the calibration already used in `GRFromMaxwell.lean`. The seed map  $\mathcal{R} \rightsquigarrow A$  of Subsection 5.3 makes the coupling explicit at the variational level: the same readout increments and  $\log(\phi + 1)$  weighting that thread into  $\mathcal{L}_\phi$  also enter the gravitational constraint through  $\phi$  and  $\alpha$ . A Tier III continuum limit would target Einstein-type dynamics with an emergent, normalization-sensitive effective vacuum contribution—not a fitted fundamental  $\Lambda$ .

**Lapse dragging.** In synchronous-comoving gauge the HQVM lapse is  $N = 1 + \Phi + \phi t/c$  (`HQVM_lapse` in `HQVMetric.lean`). Following [5, 9], we use the term lapse dragging for the coupling of the cumulative horizon phase  $\phi t/c$  to proper time and to the same  $\phi$  slot that appears in  $\mathcal{L}_\phi$  and in (4). Lean packages the identification of that slot across the *O-Maxwell* and homogeneous GR branches as `same_phi_in_O_Maxwell_and_HQVM`; the imprint exponent is the same statement as `same_alpha_in_O_Maxwell_and_HQVM`. Thus “lapse dragging” is not an extra postulate: it is the visible name for the shared  $\phi$  morphism already audited in `GRFromMaxwell.lean`.

TABLE 1. Effective gravitational coupling in natural units ( $G_0 = 1$ , `G0_eq`):  $G_{\text{eff}}(\phi)/G_0 = \phi^{3/5}$  from `G_eff_eq` (rounded).

| $\phi$ | $G_{\text{eff}}(\phi)/G_0 = \phi^{3/5}$ |
|--------|-----------------------------------------|
| 0.5    | 0.660                                   |
| 0.8    | 0.875                                   |
| 1.0    | 1.000                                   |
| 1.2    | 1.116                                   |
| 1.5    | 1.275                                   |
| 2.0    | 1.516                                   |

**Galactic rotation curves.** Flat disk kinematics in the low-acceleration regime are standard targets for modified inertia derived from horizon overlap [9]. The same  $\alpha = 3/5$  that enters  $G_{\text{eff}}(\phi) = \phi^\alpha$  and  $\mathcal{L}_\phi$  is the imprint forced by the discrete light-cone layer (`alpha_eq_3_5`); phenomenologically, a running  $G_{\text{eff}}$  tied to  $\phi$  and a Brodie-type inertia factor  $f(a_{\text{loc}}, \phi)$  are two faces of one horizon bookkeeping constraint at different kinematic scales. Table 1 makes the cosmological side of that bookkeeping numerically concrete; linking disk  $v(r)$  data to the same audited  $\alpha$  is then a Tier III continuum/stationarity task (paired Einstein and matter action stationarity in Sect. 8), not a new parameter at Tier I.

**Remark 3.5** (Superposition discipline (`action_total_general_add_J`)). Physically, adding two current systems  $J_1 + J_2$  should increase only the interaction energy  $\int J \cdot A$  (here, the discrete  $4\pi J \cdot A$  term) while retaining a single copy of the gauge-field self-energy and a single Friedmann constraint for the shared  $(\phi, \rho)$ . Equation (3) is the bookkeeping correction that enforces “no double counting” of  $\mathcal{L}_{\text{kin}}$ ,  $\mathcal{L}_\phi$ , and  $\mathcal{S}_{\text{grav}}$  when concatenating two source packages into one variational cell. This matches the broader HQIV audit discipline: only book-kept morphisms count as physical intermediates.

**Remark 3.6** (`equations_from_action`). The packaged theorem `equations_from_action` records the joint pattern: Friedmann arises as the gravitational constraint, while the gauge sector’s EL identities are the discrete equations of motion at the chosen potential.

4. HOLONOMY ALIGNMENT AND THE SAME  $F$  DATA

## 4.1. Cyclic Stokes and trivial plaquette holonomy.

**Lemma 4.1** (Cyclic flux sum (`sum_F_cyclicIndex_eq_zero`)). *For every channel  $a$  and every potential  $A$ ,*

$$\sum_{i \in \text{Fin } 4} F^a_{i, i+1}(A) = 0.$$

Expository telescoping (`sum_F_cyclicIndex_eq_zero`). Unfold  $F^a_{i, i+1} = A^a_{i+1} - A^a_i$  for  $i = 0, 1, 2, 3$  on the cyclic `Fin 4` line, with  $3 + 1 = 0$  in the index ring. The sum is

$$(A^a_1 - A^a_0) + (A^a_2 - A^a_1) + (A^a_3 - A^a_2) + (A^a_0 - A^a_3) = 0,$$

purely by cancellation; no input from octonion multiplication is used—only the abelian group structure of  $\mathbb{R}$  under addition. This is the discrete face of Stokes’ theorem on the minimal spacetime cycle at the cutoff.

**Lemma 4.2** (Trivial abelian holonomy on the cycle (`discreteSquareHolonomy_F_cyclic_eq_one`)). *With transports `linearEnd`  $(\delta)(x) = x + \delta$ , the cyclic plaquette holonomy built from edges  $F^a_{i, i+1}$  equals the identity: `discreteSquareHolonomy_F_cyclic_eq_one`.*

**Lemma 4.3** (Kinetic vs. cyclic Wilson squares (`L_0_kinetic_two_sided_cyclic_wilson_sq`)). *For any  $x \in \mathbb{R}$ , `L_0_kinetic_two_sided_cyclic_wilson_sq` gives*

$$-\frac{1}{2} \sum_{a,i} ((U^a_i x - x)^2) \leq \mathcal{L}_{\text{kin}}(A) \leq -\frac{1}{4} \sum_{a,i} ((U^a_i x - x)^2),$$

where each  $U^a_i$  is translation on  $\mathbb{R}$  by the edge flux  $F^a_{i, i+1}$  (`linearEnd` in `ActionHolonomyGlue.lean`). Thus the global variational kinetic is bounded above and below by sums of squared abelian Wilson defects on the cyclic edges.

TABLE 2. Variational vs. holonomy reading of the same  $F$  slots (`Hqiv/Physics/ActionHolonomyGlue.lean`).

| Quantity                   | Variational meaning                                            | Holonomy meaning                            | Lean identifier                                                                                                             |
|----------------------------|----------------------------------------------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| $F^a_{\mu\nu}$             | Kinetic density $-\frac{1}{4}F^2$ summed over all $(\mu, \nu)$ | Edge flux for abelian transport             | <code>F_from_A</code>                                                                                                       |
| $\sum_i F^a_{i, i+1}$      | Implied constraint on cyclic subset of pairs                   | Discrete Stokes: vanishing sum around cycle | <code>sum_F_cyclicIndex_eq_zero</code>                                                                                      |
| $\prod_i U^a_i$            | (not used directly in $L$ )                                    | Identity holonomy on closed plaquette       | <code>discreteSquareHolonomy_F_cyclic_eq_one</code>                                                                         |
| $\mathcal{L}_{\text{kin}}$ | Global $8 \times 4 \times 4$ aggregate                         | Bounded by cyclic Wilson squares            | <code>L_0_kinetic_le_neg_quarter_sum_cyclic_wilson_sq</code> ,<br><code>L_0_kinetic_ge_neg_half_sum_cyclic_wilson_sq</code> |

**Remark 4.4** ( $\text{SO}(8)$  action as internal wave-function transport). *The  $\mathfrak{so}(8)$  generators that realize the certified  $G_2 \cup \{\Delta\}$  closure act on the octonion-indexed components of the gauge potential  $A^a_\mu$*

( $a \in \text{Fin } 8$ ) precisely as infinitesimal rotations of an 8-component internal “wave function.” Left-multiplication by the imaginary units supplies the skew-adjoint operators  $L(e_i) \in \mathfrak{so}(8)$  that rotate one octonion direction into another while preserving the Euclidean norm on the carrier. The phase-lift generator  $\Delta$  then appears as a distinguished  $U(1)$ -like rotation inside this larger structure, consistent with the normalized cumulative phase readout  $\mathcal{R}$  of [1]. In this reading the abelian plaquette transports of Lemma 4.2 are literally parallel-transport operators for the internal wave function along the minimal  $\text{Fin } 4$  cycle. The interpretation is motivational and does not modify any Lean-proved statement; it simply renders manifest why the eight-dimensional normed division algebra is the natural carrier once phase information and curvature are already present in the HQIV pipeline.

Physical interpretation. The HQIV null-shell law is quadratic in the shell index while coarse spatial ball counting is cubic [2]; the curvature channel built on that law feeds  $\alpha$  and hence both monogamy bookkeeping and the variational  $\phi$ -coupling (Remark in Sect. 2). Table 2 makes the parallel explicit at the *field* level: the same discrete  $F$  enters a global variational kinetic and a local Wilson/Stokes picture, so coercive control on plaquette defects is inherited from the same object that defines the action—the signature that the cutoff is not an arbitrary bookkeeping choice split across modules. Non-abelian upgrade (pointer). Matrix-valued transports inside `Function.End` for the same combinatorial cycle are packaged in `Hqiv/Physics/DiscretePlaquetteHolonamy.lean`; the glue file stays in the abelian  $\mathbb{R}$  sector to isolate the discrete Stokes identity.

## 5. DESIGN BRIDGE: FROM GROWTH READOUT TO GAUGE VARIABLES

### 5.1. Shared parameters.

- **Curvature channel**  $\Rightarrow \alpha$ . The discrete densities  $\rho$ ,  $K$ ,  $\Omega$  and the imprint  $\alpha$  are defined in `OctonionicLightCone / HQVMetric`; the same  $\alpha$  appears in  $\mathcal{L}_\phi$  and in  $G_{\text{eff}}(\phi) = \phi^\alpha$  (`G_eff_eq`).
- **Phase readout**  $\Rightarrow \Delta$  and  $\mathfrak{so}(8)$ . [1] constructs  $\Delta$  from the normalized readout; closure spans the 28-dimensional Lie algebra on the HQVM seed.
- **Auxiliary  $\phi$** . `AuxiliaryField` ties  $\phi(m)$  to the temperature ladder and hence to the same horizon bookkeeping used in the curvature numerator.
- **EL  $\Leftrightarrow$  emergent slot**. Stationarity in  $A$  is aligned with the emergent Maxwell source pattern (Remark 2.6).

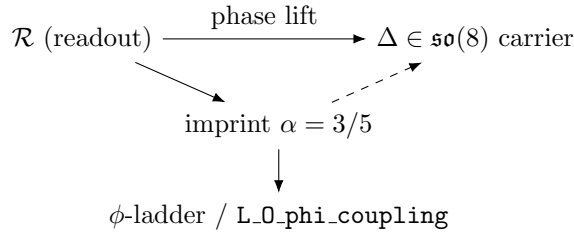


FIGURE 1. Design bridge (schematic): the readout pipeline fixes  $\Delta$  in the skew-adjoint octonionic carrier while the curvature imprint  $\alpha$  feeds the auxiliary-field ladder and the action’s  $\phi$ -coupling. A fully functorial arrow  $\mathcal{R} \rightsquigarrow A$  is not yet formalized (see `ManifoldLagrangianScaffold.lean`); Subsection 5.3 gives an explicit Stokes-compatible seed on the minimal  $\text{Fin } 4$  cycle.

**5.2. Functoriality gap.** A morphism  $\mathcal{R} \rightsquigarrow A$  with estimates in a mesh-refinement limit is explicitly deferred to `ManifoldLagrangianScaffold.lean` (`LagrangianDensity`, chart pullbacks) and the coincidence hypothesis `LatticeContinuumActionCoincidence`. Subsection 5.3 gives the reference-plaquette seed, now implemented as `seedPotentialMinimalCycle` and `imprintWeightedReadoutPhase` in `Hqiv/Physics/ReadoutGaugeSeed.lean`; functorial extension to larger lattices and error-controlled continuum charts remain Tier III (Sect. 8).

**5.3. Minimal seed map  $\mathcal{R} \rightsquigarrow A$  on the reference Fin4 cycle.** This subsection packages a single, fully explicit assignment from the normalized readout and auxiliary ladder to edge fluxes on the smallest nontrivial spacetime cycle, compatible with Lemma 4.1 and with the abelian holonomy lemmas of Sect. 4. The same assignment is now defined and certified in `Hqiv/Physics/ReadoutGaugeSeed.lean`. It is the natural first discrete closure of Figure 1 on one plaquette before any continuum limit.

Setup (Lean-identified quantities). Fix the calibration shell `referenceM` (definitionally tied to the QCD-onset ladder in `OctonionicLightCone.lean`; numerically 4 in the current repository constants). Under positivity of `curvature_integralreferenceM`, the partial curvature ratio satisfies `omega_k_partialreferenceM=1` (`omega_k_partial_at_reference`)—the discrete normalization pin for the  $\Omega$  layer in [1]. Write  $\mathcal{R}(n)$  for the normalized cumulative readout (the  $\mathcal{R}$  of [1], implemented via the  $\Omega$  pipeline) at shell  $n$ , and  $\phi(n)$  for `phi_of_shell n` from `AuxiliaryField`. Define the per-shell readout increment

$$\delta_{\mathcal{R}}(n) := \mathcal{R}(n+1) - \mathcal{R}(n),$$

and the imprint-weighted phase increment

$$(5) \quad \omega(n) := \alpha \log(\phi(n) + 1) \delta_{\mathcal{R}}(n),$$

with  $\alpha = 3/5$  as in `alpha_eq_3_5`. The phase-lift generator  $\Delta \in \mathfrak{so}(8)$  is fixed in the  $(e_1, e_7)$  plane in the HQVM conventions of [1]; let  $\theta$  denote the corresponding polar angle in that plane and set

$$v^1 := \cos \theta, \quad v^7 := \sin \theta, \quad v^a := 0 \quad (a \notin \{1, 7\}),$$

so  $v \in \mathbb{R}^8$  is a unit vector in the  $\Delta$ -direction.

Stokes-compatible flux on the Fin4 cycle. For a *fixed* shell label  $n$ , assign internal components on directed edges  $i \rightarrow i+1$  of the cyclic Fin4 graph by

$$(6) \quad F_{i,i+1}^1(n) = \omega(n) s_i \cos \theta, \quad F_{i,i+1}^7(n) = \omega(n) s_i \sin \theta, \quad F_{i,i+1}^a(n) = 0 \quad (a \notin \{1, 7\}),$$

where  $(s_0, s_1, s_2, s_3) = (+1, +1, -1, -1)$ . Then  $\sum_{i \in \text{Fin4}} F_{i,i+1}^a(n) = 0$  for every  $a$  and every  $n$ , so Lemma 4.1 holds *by construction*. Recover potentials on vertices by fixing  $A_0^a(n) = 0$  and defining

$$A_{i+1}^a(n) := A_i^a(n) + F_{i,i+1}^a(n) \quad (i = 0, 1, 2, 3),$$

with indices interpreted cyclically so that  $A_4^a(n) = A_0^a(n)$ ; this is the standard pure-gauge freedom on a loop: only the edge data enter `F_from_A`, `L_0_kinetic`, and the holonomy glue.

Verification (three checks).

- (1) **Cyclic flux (Lemma 4.1).** The choice (6) forces  $\sum_i F_{i,i+1}^a(n) = 0$  by the alternating signs; equivalently, the vertex potential closes after four steps. No octonion multiplication is used—only  $\mathbb{R}$ -addition, matching the proof style of `sum_F_cyclicIndex_eq_zero`.

- (2) **Abelian holonomy (Lemma 4.2).** In `ActionHolonomyGlue.lean`, cyclic transport is built from `linearEnd` applied to the edge variables  $F_{i,i+1}^a$ ; Lemma 4.2 gives the identity holonomy whenever the cyclic sum vanishes. Thus the seed is automatically compatible with the proved  $\mathbb{R}$ -translation picture. An *interpretive* upgrade identifies  $\omega(n)v$  with an infinitesimal rotation in the  $\Delta$ -plane inside  $\mathfrak{so}(8)$ ; the exponential  $\exp(\cdot \Delta)$  language belongs to that matrix sector and is not literal `linearEnd` composition until a non-abelian transport layer is wired (Sect. 4, non-abelian pointer).
- (3) **Reference normalization.** At `referenceM`,  $\omega(n)$  vanishes whenever  $\delta_{\mathcal{R}}(n) = 0$  at that shell (e.g. a bookkeeping pivot for the readout increment); then (6) collapses to  $F = 0$  and, with  $A(\cdot, n) \equiv 0$  at the base vertex, to the flat cycle used in Example 3.3. More generally,  $\omega(n)$  couples the curvature channel through  $\delta_{\mathcal{R}}$  and the auxiliary factor  $\log(\phi(n) + 1)$  that already appears in  $\mathcal{L}_\phi$  (Sect. 2); the closed form  $\phi(n) = 2(n + 1)$  from `phi_of_shell_closed_form` is independent of the Friedmann  $\phi_{\text{val}}$  normalization in the action cell unless explicitly identified.

Status and scope. `Hqiv/Physics/ReadoutGaugeSeed.lean` defines `seedAProfileAux` (the  $0, \omega c, \omega c, 0$  vertex profile equivalent to the alternating edge pattern (6)), the gauge potential `seedPotentialMinimalCycle`, and the readout-linked scalar `imprintWeightedReadoutPhase`. Theorems `seedPotentialMinimalCycle_cyclic_sum_F` and `seedPotentialMinimalCycle_discrete_holonomy_one` specialize `sum_F_cyclicIndex_eq_zero` and `discreteSquareHolonomy_F_cyclic_eq_one`; `seedPotentialMinimalCycle_of_imprint_increment_zero` encodes flat degeneration when `omega_k_partial` has vanishing shell-to-shell increment. What remains outside Lean is promoting shell-indexed potentials  $A(\cdot, n)$  to a global variational field in `Action.lean`, proving functoriality and mesh-refinement estimates, and upgrading abelian `linearEnd` transport to matrix  $\exp(\cdot \Delta)$  data—the Tier III program in Sect. 8.

## 6. TIERED UNIQUENESS THESIS

**6.1. Tier I: EL slot rigidity (formal layer).** In continuum polynomial abelian gauge theory,  $\mathcal{L} \sim -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + J_\mu A^\mu$  yields  $\partial_\mu F^{\mu\nu} = 4\pi J^\nu$  up to gauge-fixing nuances. The discrete HQIV choice replaces  $\partial_\mu$  by the finite-difference operator whose quadratic form is exactly (1); Theorem 2.4 is therefore the *same* uniqueness of functional derivative up to the declared  $J$  and  $\phi$  slots—proved in Lean by unfolding definitions (`EL_0_general_eq_F_divergence_sub_sources`, `action_0_Maxwell_EL_eq_emergent_general`).

Finite-difference dictionary. If one thinks of  $\mu \in \text{Fin } 4$  as labeling vertices of a periodic one-cell graph, then  $F_{\mu\nu}^a$  is the oriented edge jump from  $\mu$  to  $\nu$ . The EL expression  $\sum_\mu F_{\mu\nu}^a$  is the net flux out of vertex  $\nu$  after summing over all partners  $\mu$ : the same combinatorial object that becomes  $\partial_\mu F^{\mu\nu}$  after mesh refinement under uniform bounds on  $A$ . Thus Tier I is not an *ad hoc* discrete trick; it is the standard Maxwell EL with honest finite differences on the HQIV index set.

What is *not* determined at Tier I.. Adding total “discrete divergences” that vanish identically on every  $A$  would change  $\mathcal{L}$  by a null Lagrangian without changing EL; the HQIV file fixes a definite representative (the normalized  $F^2$  sum (1)). Uniqueness claims at Tier I are therefore uniqueness of the *chosen* representative’s EL slot, not uniqueness of all possible discrete prelimits of the same continuum theory.

**6.2. Tier II: division algebra and eight channels (classical layer).** Hurwitz’s theorem classifies finite-dimensional normed division  $\mathbb{R}$ -algebras as  $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$  [7, 8]. Baez emphasizes the “normed

division algebra ladder” stopping at  $\mathbb{O}$ ; Springer–Veldkamp supply the standard structure theory (alternative algebras, norm forms,  $\text{Aut}(\mathbb{O}) = G_2$ ). Only  $\mathbb{O}$  has real dimension 8, matching the Fin8 carrier and the Spin(8) triality story reviewed in [2]. Once the HQVM matrices and  $\Delta$  are fixed, the machine-checked closure  $G_2 \cup \{\Delta\} \Rightarrow \mathfrak{so}(8)$  [1] pins an exceptional 28-dimensional bracket envelope on that carrier. Tier II uniqueness is therefore *algebraic compatibility*: the octonions are the unique normed division algebra of the dimension required for this particular closure package. Triality as a selection rule (informal). Among division algebras, only the 8-dimensional case carries outer automorphisms of Spin(8) large enough to coordinate vector, spinor, and conjugate-spinor roles in the way the HQIV narrative packages internal quantum numbers. The present paper does not reprove triality; it records that the Lean closure certificate is *specialized* to the octonionic  $8 \times 8$  seed, so any alternate division-algebra carrier would require a fresh closure script of comparable strength.

**6.3. Tier III: program (continuum apex).** We isolate three explicit open sub-problems rather than a single vague claim:

- (1) **Discrete–continuum limit.** Construct controlled morphisms from  $\mathcal{R}$  (and shell ledgers) to connection/torsion data with error bounds tied to the quadratic null law.
- (2) **Paired stationarity.** Promote  $\mathcal{S}_{\text{tot}}$  to a manifold integral whose Euler–Lagrange system recovers paired Einstein-type and  $G_2$  holonomy constraints in a sharp asymptotic regime.
- (3) **Associator/measure.** Encode the octonionic associator  $(x, y, z)$  in the discrete measure or boundary term so quaternionic/sedenionic truncations fail structurally, and link to a Tier III path integral.

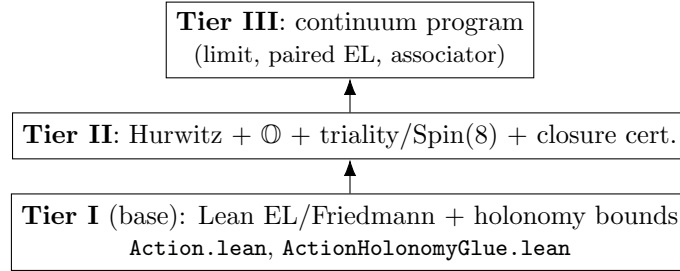


FIGURE 2. Tiered uniqueness pyramid: formal theorems at the base, classical rigidity in the middle, conjectural continuum completion at the apex. Cosmology-facing consequences of Tier I (Friedmann constraint,  $G_{\text{eff}}$ , lapse dragging) are summarized in Remark 3.4 and Table 1.

## 7. QUANTUM SCAFFOLD AND FORWARD COMPATIBILITY

`Hqiv/QuantumMechanics/Schrodinger.lean` defines `eulerLagrange_eq_Schrodinger` and proves `actionExtensionYieldsSchrodinger`: the extended-action EL predicate is definitionally the Schrödinger scaffold with the HQIV Hamiltonian template. Placeholder Laplacians remain; when Tier III supplies a continuum measure on fields, this module is the natural place to assert square-integrability and variational well-posedness compatible with that measure, including BRST or BV upgrades if the associator enters as a constraint closing ghost sectors.

## 8. OUTLOOK AND ROADMAP FOR TIER III

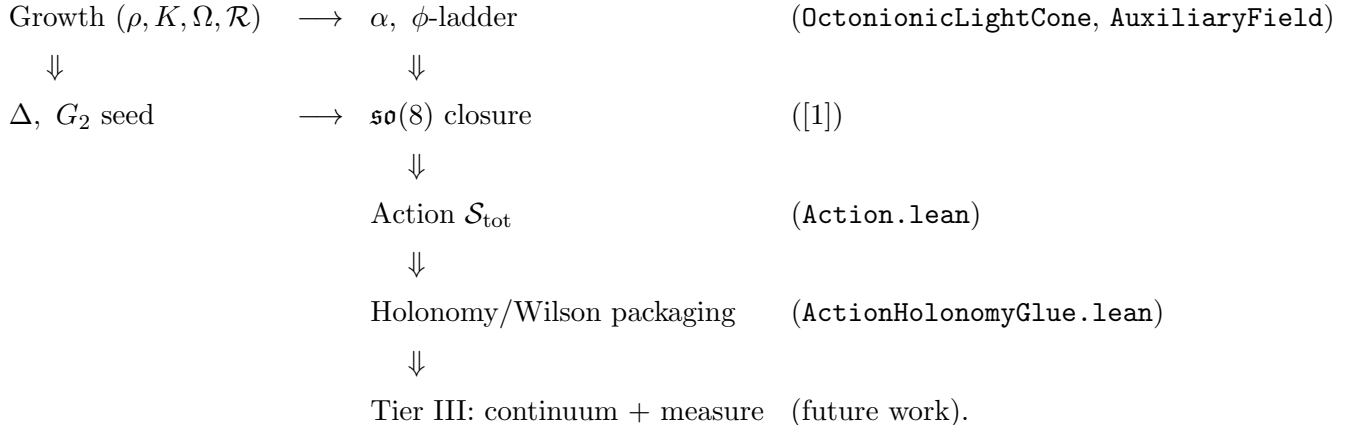
## 8.1. Minimal additional Lean scaffolds.

- **LatticeContinuumActionCoincidence**: hypothesis interface matching discrete sums to continuum integrals.
- Controlled *field* bridge lemmas  $\mathcal{R} \rightsquigarrow A$  (or  $\Delta \rightsquigarrow A$ ) with explicit error terms, building on **ManifoldLagrangianScaffold**.
- Discrete measure or boundary term slot for the octonionic associator, and quaternionic/sedenionic obstruction lemmas stated as **false** targets under the same ansatz.
- Wiring **coordsGradientComponents** fully through both **EL\_0\_general\_coordsField** and **emergentMaxwellInhomogeneous\_0\_coordsField** to close the Action  $\leftrightarrow$  ModifiedMaxwell loop.

## 8.2. Milestone ordering (suggested).

- (1) Close the Action  $\leftrightarrow$  ModifiedMaxwell gradient identification on charts (**ContinuumOmaxwellClosure**).
- (2) Promote **LatticeContinuumActionCoincidence** from a hypothesis structure to a theorem template with explicit rate bounds.
- (3) Build a functor  $\mathcal{R} \rightsquigarrow A$  on a restricted class of asymptotically HQIV configurations (partial maps first, estimates second).
- (4) Add non-abelian holonomy lemmas mirroring **L\_0\_kinetic\_two\_sided\_cyclic\_wilson\_sq** for matrix transports in **DiscretePlaquetteHolonomy**.
- (5) Encode associator weights in a discrete path measure; prove quaternionic obstruction lemmas.
- (6) Revisit **Schrodinger.lean** once measure is fixed, upgrading placeholders to **deriv** along certified charts.

## 8.3. Dependency sketch (text).



**8.4. Thermodynamic root and lattice gauge vocabulary.** Jacobson’s thermodynamic state relation for spacetime [5] is the standard conceptual ancestor of “Einstein equation of state” arguments; Brodie’s horizon package [9] is the imported normalization route for  $\alpha = 3/5$  in the HQIV growth papers and feeds the same imprint into  $\mathcal{L}_\phi$  and  $G_{\text{eff}}$  here. Separately, Wilson’s lattice link variables [4] and the textbook discrete gauge theory framework [6] supply the vocabulary for reading Lemma 4.3 as a small-flux plaquette inequality: the HQIV glue lemmas are not a new

lattice gauge theory, but they position the O–Maxwell kinetic where Wilson-type estimates can be imported when Tier III tightens scalings.

#### APPENDIX A. LEAN IDENTIFIER INDEX (SELECTED)

Table 3 collects identifiers used repeatedly in this manuscript; it is a reader map, not an exhaustive dependency graph (see [3]).

TABLE 3. Selected Lean identifiers for the variational layer.

| Identifier                                                                                                         | Role                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>F_from_A</code>                                                                                              | Discrete field strength $A^a_\nu - A^a_\mu$ .                                                                                                                      |
| <code>L_0_kinetic</code> , <code>L_0_source_</code><br><code>general</code> , <code>L_0_phi_coupling</code>        | Kinetic, source, $\phi$ terms.                                                                                                                                     |
| <code>L_0_Maxwell_general</code> , <code>action_</code><br><code>O_Maxwell_general</code>                          | Full O–Maxwell density / cell action.                                                                                                                              |
| <code>EL_0_general</code> , <code>action_0_</code><br><code>Maxwell_EL_eq_emergent_</code><br><code>general</code> | EL covector; equality to divergence form.                                                                                                                          |
| <code>action_0_Maxwell_general_</code><br><code>add_J</code>                                                       | Nonlinear superposition correction for shared kinetic/ $\phi$ pieces.                                                                                              |
| <code>S_HQVM_grav</code> , <code>S_HQVM_grav_</code><br><code>zero_iff_Friedmann</code>                            | Gravitational constraint scalar; Friedmann iff zero.                                                                                                               |
| <code>action_total_general</code> ,<br><code>action_total_general_add_J</code>                                     | Total cell; corrected sum of two currents.                                                                                                                         |
| <code>sum_F_cyclicIndex_eq_zero</code> ,<br><code>discreteSquareHolonomy_F_</code><br><code>cyclic_eq_one</code>   | Discrete Stokes; trivial cyclic holonomy.                                                                                                                          |
| <code>L_0_kinetic_two_sided_</code><br><code>cyclic_wilson_sq</code>                                               | Bounds kinetic by cyclic Wilson squares.                                                                                                                           |
| <code>phi_of_shell</code> , <code>shell_shape_</code><br><code>in_terms_of_phi</code>                              | Auxiliary field ladder tied to temperature shells.                                                                                                                 |
| <code>G_eff_eq</code> , <code>three_minus_gamma_</code><br><code>eq</code>                                         | Varying $G$ ; Friedmann prefactor $(3 - \gamma)$ .                                                                                                                 |
| <code>HQVM_lapse</code> , <code>G0_eq</code> , <code>same_phi_</code><br><code>in_0_Maxwell_and_HQVM</code>        | ADM lapse $N = 1 + \Phi + \phi t/c$ ; natural-unit $G_0$ ; shared $\phi$ for lapse/time-angle and auxiliary ladder ( <i>lapse dragging</i> ; Remark 3.4, Table 1). |

#### APPENDIX B. CROSS-SECTOR COMPATIBILITY (`GRFROMMAXWELL.LEAN`)

Remark 3.4 gives the expository name *lapse dragging* for the morphism packaged below. The variational story is intentionally not isolated from the metric sector: `Hqiv/Physics/GRFromMaxwell.lean` packages a sequence of equalities asserting that the same horizon fields drive O–Maxwell and HQVM-GR bookkeeping in the homogeneous limit. Representative lemmas include `same_phi_in_0_Maxwell_and_HQVM` (lapse/time-angle uses the same  $\phi$  as the auxiliary ladder), `same_alpha_in_0_Maxwell_and_HQVM` (the lattice imprint matches the exponent in  $G_{\text{eff}}(\phi) = \phi^\alpha$ ), and `O_Maxwell_compatible_with_HQVM_GR_homogeneous` (Friedmann as the explicit (13/5) rational form when  $\phi \geq 0$ ). These files are the correct place to cite when a referee asks whether  $\mathcal{L}_\phi$  “knows” about cosmology: it is the same  $\alpha$  and  $\phi$  pipeline, not a second independent tuning knob.

## ACKNOWLEDGMENTS

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